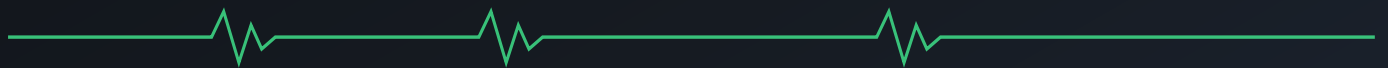


Putting TRIAGE-LINK on Responder Devices

Install the offline-first PWA on Android phones & tablets, Windows Surface & laptops, and iPad — by hand or across a managed fleet. Companion to the Technical Architecture (§13 deployment topology, §14 security).



Document v0.1 Scope Device install + MDM Audience Ops · IT · Field leads

△ Prototype – installs the current build as an app. Add the §14 security controls before entering real patient data.

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TRIAGE-LINK is an **offline-first Progressive Web App** — there is no app-store binary to sideload. Deployment is three steps: **build & host** the app once on an HTTPS URL, **install** it on each device (one tap — it pins like a native app), then run it **fully offline** in the field. The network is only needed again to receive updates.

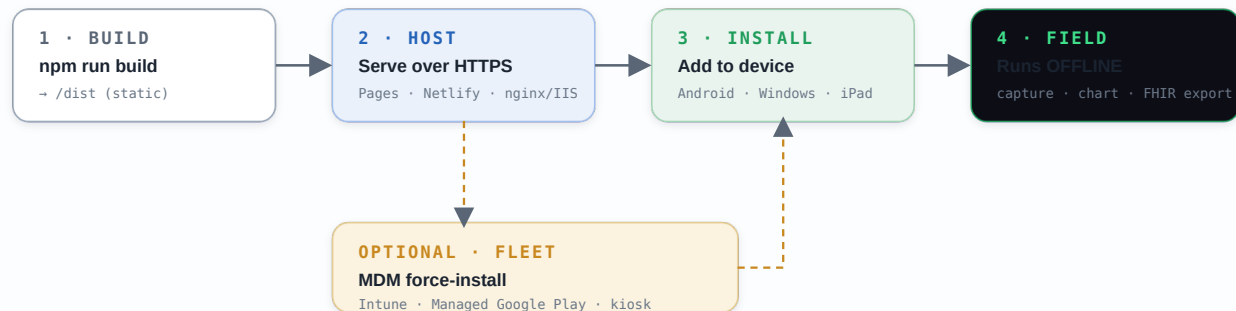


FIG 1 – Build once, host on HTTPS, install per device, run offline. For a fleet, MDM installs the same hosted URL automatically.

01 Prerequisites

REQUIREMENT	WHY
Served over HTTPS	PWAs install and run a service worker only in a secure context . <code>https://...</code> works; a plain LAN address like <code>http://192.168.x.x</code> does not . <code>localhost</code> / <code>127.0.0.1</code> are exempt, for testing only.
A modern browser	Android → Chrome (or Samsung Internet); Windows → Edge or Chrome ; iPhone/iPad → Safari . Keep them updated.
~50-100 MB free storage	For the cached app shell and the local IndexedDB record store.
The built /dist folder	Produced by <code>npm run build</code> — this is what you host.

Build the app

```

npm install
npm run build      # type-checks, then emits the static site to /dist
npm run preview    # optional: serve /dist locally at http://localhost:4173 to test

```

`/dist` is a self-contained static bundle — no server runtime, no database. Whatever hosts it only needs to serve static files over HTTPS.

02 Host the app (one-time)

Pick whichever fits your situation — all that matters is a valid HTTPS URL.

OPTION	GOOD FOR	NOTES
<code>npm run preview</code>	Quick local test on the same machine	<code>localhost</code> is a secure context, so install works for testing
GitHub Pages	Free public demo of the prototype	Project sites serve under a sub-path — see the base-path note. Don't put real PHI on a public host.
Netlify / Vercel	Free hosting, automatic HTTPS	Drag-and-drop <code>/dist</code> or connect the repo
Internal nginx / IIS	Real field/enterprise use behind your firewall	Serve <code>/dist</code> over HTTPS with a cert your devices trust (public or org CA)

i Base-path note (GitHub Pages & sub-path hosting). If you host at `https://<user>.github.io/Emergency-Medical-Support-System/` instead of a domain root, set the matching base in `vite.config.ts` so the service-worker scope and asset paths resolve, then rebuild:

```
export default defineConfig({
  base: '/Emergency-Medical-Support-System/',
  // ...existing plugins
})
```

At a domain root (Netlify / Vercel / your own domain) leave `base` as the default `('/')`.

03 Android phones & tablets (Chrome)

- 1 Open the hosted **HTTPS URL** in **Chrome**.
- 2 Let it finish loading once — this lets the service worker cache the app for offline use.
- 3 Tap **⋮** (top-right) → **Install app** (or **Add to Home screen**). Chrome often shows an **Install** banner automatically once the page qualifies.
- 4 Confirm. A TRIAGE-LINK icon appears on the home screen / app drawer.
- 5 Launch from that icon — it opens full-screen, without browser chrome, like a native app.

Samsung Internet: menu → **Add page to** → **Home screen** works the same way.

- ✓ **Verify offline:** turn on **Airplane mode**, open the app from its icon, and confirm you can create a casualty, place injuries, record vitals, and export a FHIR bundle.

04 Windows Surface & laptops (Edge)

Microsoft Edge (recommended on Surface):

- 1 Open the **HTTPS URL** in Edge and let it load once.
- 2 Click the **install icon** in the address bar (a monitor with a ↓), or **⋮** → **Apps** → **Install this site as an app**.
- 3 Click **Install**; pin to **Start** / **Taskbar** when prompted.
- 4 It launches in its own window and appears in the Start menu like any installed app.

Google Chrome on Windows: click the **install icon** in the address bar → **Install**.

Surface specifics: touch and the Surface Pen both work on the body chart exactly as a mouse does — tap an injury type, then tap the body to drop a marker. No extra setup.

- ✓ **Verify offline:** disconnect Wi-Fi, then launch the app from the Start menu and run through a capture.

05 iPhone & iPad (Safari)

- 1 Open the **HTTPS URL** in **Safari** (PWA install on iOS is Safari-only).
- 2 Tap **Share** → **Add to Home Screen** → **Add**.
- 3 Launch from the new home-screen icon; it runs full-screen.

! **iOS caveats:** there's no automatic install prompt, and iOS may evict cached data if the device is very low on storage and the app is unused for a long time. For a managed fleet, Android or Windows give firmer control.

06 Fleet / MDM deployment

For more than a handful of devices, push the app from your device manager so every responder tablet gets it — and the right security policy — automatically.

Android fleet — Intune / Workspace ONE / SOTI + Managed Google Play

- ▶ Use **Managed Google Play** → **"Web apps"** to wrap the hosted URL as an installable web app (Google builds a signed WebAPK). Assign it to your responder group as a **required** app so it force-installs.
- ▶ For single-purpose tablets, use **dedicated-device / kiosk mode** to lock the device to TRIAGE-LINK alone.
- ▶ Enforce: **device encryption**, **screen lock / biometric**, **auto-update**, and **remote wipe**.

Windows / Surface fleet — Microsoft Intune

- ▶ Edge supports a **force-install policy for web apps** (`WebAppInstallForceList`). Point it at the hosted URL and Edge installs + pins TRIAGE-LINK on managed devices at next launch — no user action.
- ▶ Pair with **BitLocker** (disk encryption), **Windows Hello** (biometric sign-in), and, for dedicated Surfaces, **Assigned Access / kiosk**.

i Exact menu paths in Intune / Managed Google Play change over time — treat the above as the mechanism and confirm current steps in your MDM console's documentation. All you ever hand the MDM is the **hosted HTTPS URL**; it does the rest.

07 Updates

Because it's a PWA you **never repackaging or re-push** to update — just re-deploy `/dist` to the host:

- 1 Make changes, run `npm run build`, upload the new `/dist` to the same URL.
- 2 Next time a device opens the app **with connectivity**, the service worker fetches the new version in the background.
- 3 Depending on the `vite-plugin-pwa` `registerType`, the update applies automatically on next launch (**autoUpdate**) or shows a "new version available" prompt (**prompt**). A quick close-and-reopen guarantees the new version is active.

Devices that are offline keep running the last cached version safely and pick up the update whenever they next reconnect.

08 Offline-readiness checklist

Run this on **each device** before it goes into the field:

- ☐ App installed from the HTTPS URL and pinned to the home screen / Start menu
- ☐ Opened **once online** so the service worker cached the app shell
- ☐ **Airplane mode on** → app still launches from its icon
- ☐ Can create a casualty, place injuries, and record vitals offline
- ☐ **Export FHIR** produces a `...-fhir-bundle.json` download on-device
- ☐ Close and reopen → the saved casualty is still there (IndexedDB persisted)
- ☐ Device encryption, screen lock, and (if managed) MDM enrolment are on

09 Device hardening (required before real PHI)

These are **operating-system / MDM** controls, not app features — but they are what makes a device safe to hold patient data. They map to `ARCHITECTURE.md` §14.

CONTROL	ANDROID	WINDOWS / SURFACE	IPAD
Encryption at rest	File-Based Encryption (default)	BitLocker	Data Protection (default)
Screen lock / biometric	PIN + fingerprint/face	Windows Hello	Passcode + Face/Touch ID
Remote wipe	via MDM	via Intune	via MDM
Auto-update OS & browser	Managed Google Play	Intune / Windows Update	MDM
Single-app lock (optional)	Kiosk / dedicated device	Assigned Access kiosk	Guided Access / kiosk

10 Quick reference

DEVICE	BROWSER	INSTALL PATH	OFFLINE AFTER INSTALL
Android phone / tablet	Chrome	⋮ → Install app	Yes
Windows Surface / laptop	Edge	⋯ → Apps → Install this site as an app	Yes
Windows laptop	Chrome	Address-bar install icon	Yes
iPhone / iPad	Safari	Share → Add to Home Screen	Yes (iOS storage caveats)

! The one rule that catches everyone: the app must be reached over **HTTPS** (or `localhost`) for installation to be offered. Plain `http://` LAN addresses load the page but won't let it install or work offline.

TRIAGE-LINK · Field Deployment Guide · v0.1 – Prototype reference, not for clinical use.
Companion to the Technical Architecture (§13 deployment topology, §14 security). Add encryption, authentication and audit before entering real patient data.